

## Trainings ▾

### General Information

The Holset® SS191B air compressor is an engine-driven, piston-type compressor that supplies compressed air to operate air-activated devices. The compressor operates continuously but has a "loaded" and "unloaded" operating mode. The operating mode is controlled by a pressure activated governor and the compressor unloading assembly. When the air system reaches a predetermined pressure, the governor applies an air signal to the air compressor unloader assembly, causing the unloader valve to hold the compressor intake valve open and compressed air to stop flowing into the air system. As the air in the air system is used, the pressure drops. At a predetermined pressure, the governor exhausts the air signal to the compressor unloader assembly, allowing the compressor to again pump compressed air into the air system.

This air compressor is available in an air cooled or water cooled version, the **only** difference being changes to the cylinder head to incorporate a water passage.

This air compressor is also available with or without a power steering pump housing.

Other brands of compressors can be used on the B Series engine. Troubleshooting procedures are very similar for these air compressors compared to the Holset® SS191. Refer to the specific air compressor manufacturer's manual for detailed repair information, including torque values.

**Note :** The cylinder head and unloader components of the Holset® SS191 air compressor can be serviced without removal of the air compressor on many engine applications. This troubleshooting manual will cover servicing of the air compressor unloader components while the compressor is still on the engine. All other servicing of the air compressor internal components should be done after the compressor has been removed from the engine. Reference the Shop Manual, Bulletin 3810433, for detailed disassembly and assembly information on air compressor SS191.

The Holset® SS296 single-cylinder air compressor is an engine-driven, piston-type compressor that supplies compressed air to operate air-activated devices. The compressor runs continuously but has a loaded and unloaded operating mode. The operating mode is controlled by a pressure-activated governor and the compressor unloading assembly.

The SS296 air compressor used on B Series engines uses an (E-type) unloader. The economy (E-type) unloader system was designed to reduce pumping losses and engine boost pressure losses through the compressor intake valve while operating in unloaded mode.

When the air system reaches a predetermined pressure, the governor applies an air signal to the air compressor unloader assembly, causing the unloader cap to seal off incoming air at the intake valve and compressed air to stop flowing into the air system.

**Note :** System pressure must be maintained on the outlet side of the discharge valve to keep the discharge valve closed.

As the air in the air system is used, the pressure drops. At a predetermined pressure, the governor exhausts the air signal to the compressor unloader assembly, allowing the compressor to again pump compressed air into the air system.

If the air system pressure is **not** maintained on the discharge valve during unloaded operation, air will be pumped out of the compressor cylinder, causing a low-pressure (vacuum) condition to form in the cylinder. With the intake valve sealed off by the unloader cap and the exhaust valve being a one-way pressure actuated valve, no air will be allowed to enter the cylinder. When the compressor cylinder pressure falls below crankcase pressure, oil will be drawn past the piston rings and pumped into the air system.

Other brands of compressors can be used on B Series engines. Troubleshooting procedures are very similar for these air compressors compared to the Holset® SS296. Refer to the specific air compressor manufacturer's manual for detailed repair information and torque specifications.